Semantic Segmentation of Indoor Scenes

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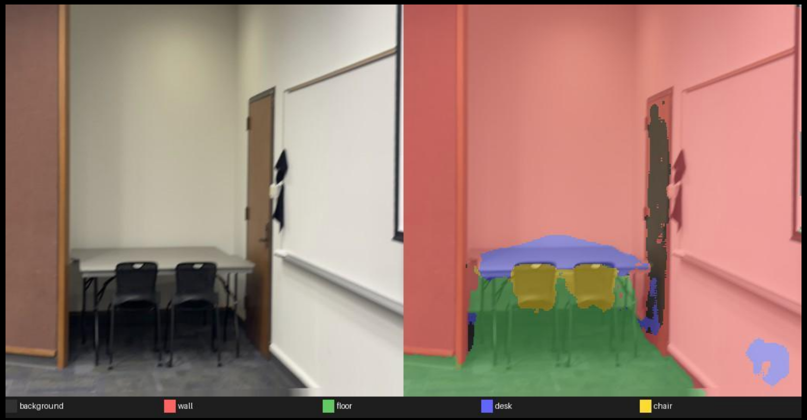
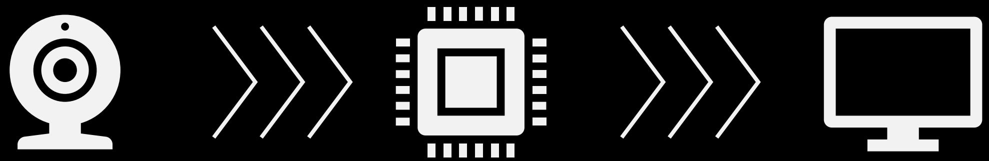
Problem & Motivation -> Approach

- Our goal is to label every pixel in an image with the use of semantic segmentation
- This will help identify indoor objects such as:
  - Walls
  - Floors/Ceilings
  - Tables
  - Chairs
- The goal is to decompose a complex environment into simple areas which can be used by another system.

- Capture RGB camera input.
- Resize and normalize the frame.
- Run a U-Net segmentation model.
- Overlay the predicted mask in real time.

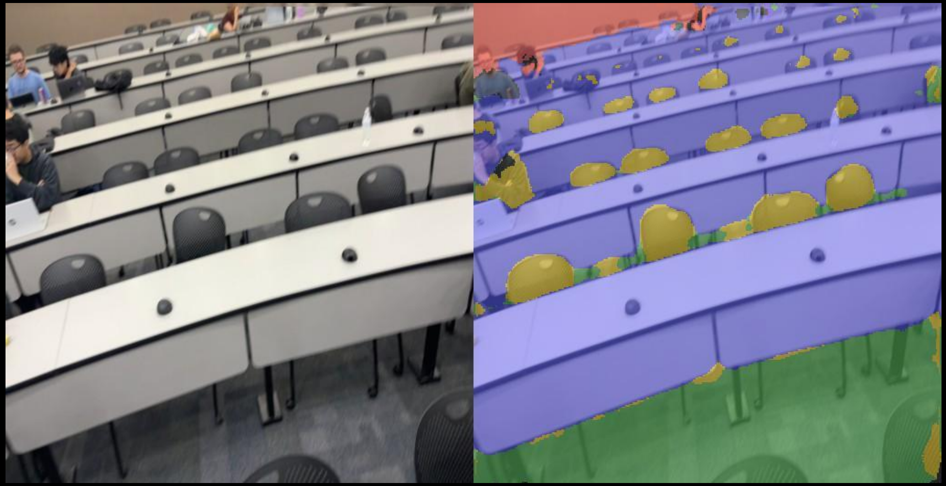


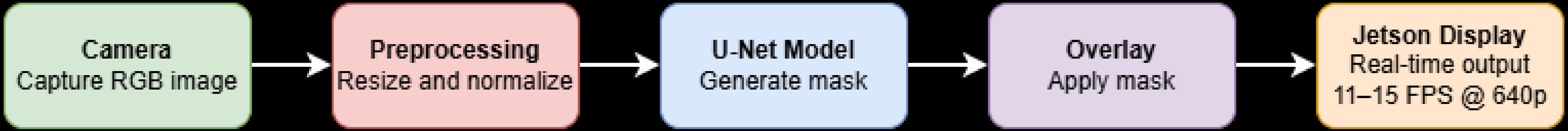


System & Hardware

- Jetson Orin Nano, the embedded deployment device
- USB webcam: live RGB input @ 480 × 640
- PyTorch: model training and testing
- ONNX/TensorRT: this is the optimized inference path
- OpenCV: camera capture and visualization
- Target: real-time segmentation at camera native resolution (480 × 640)

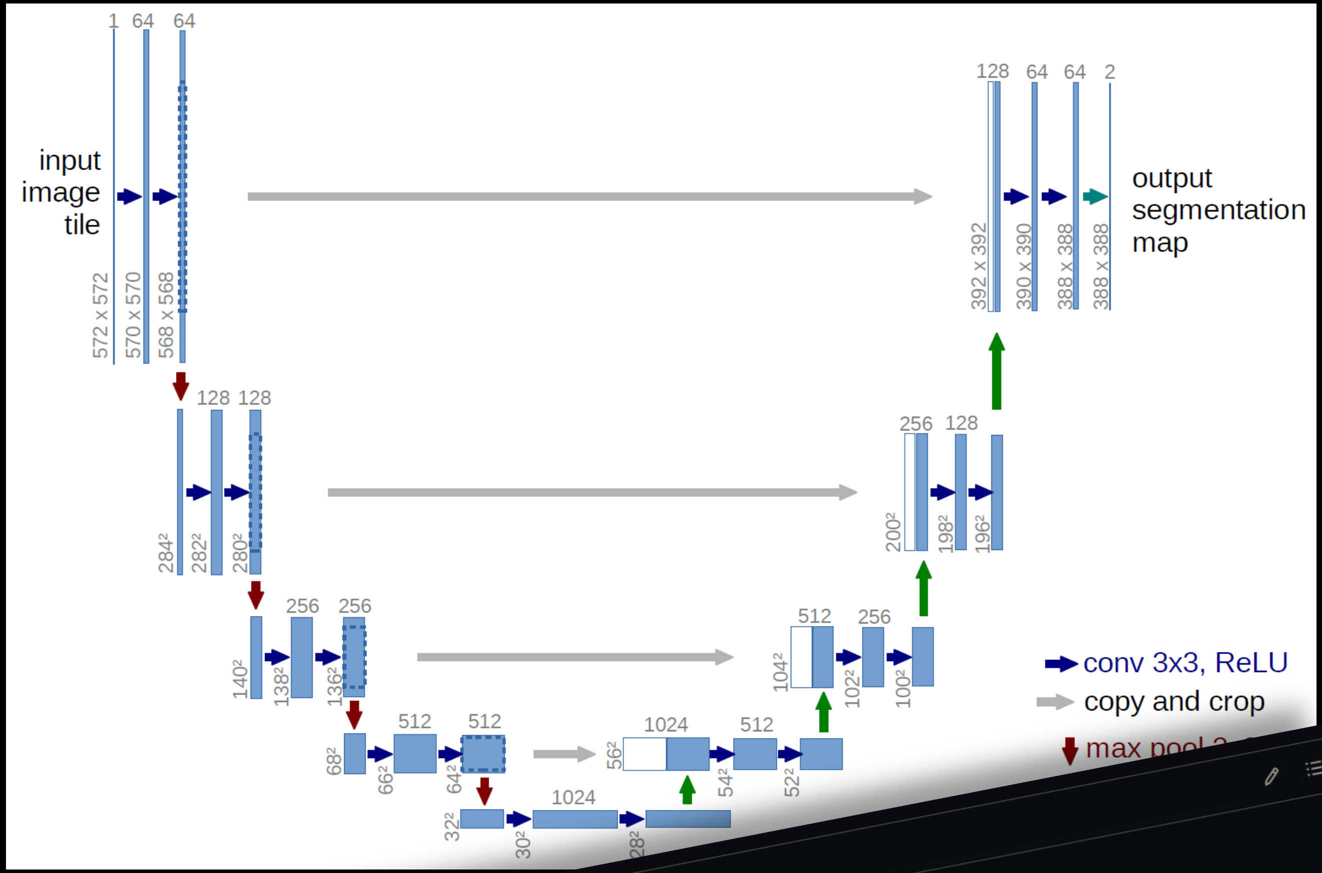
System Pipeline

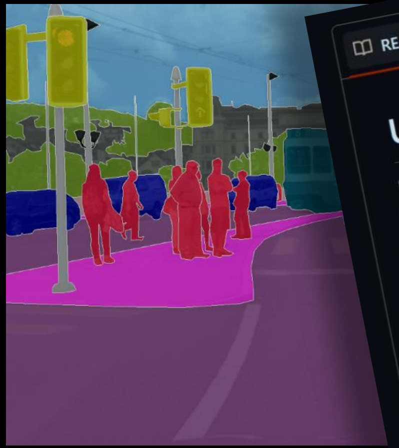


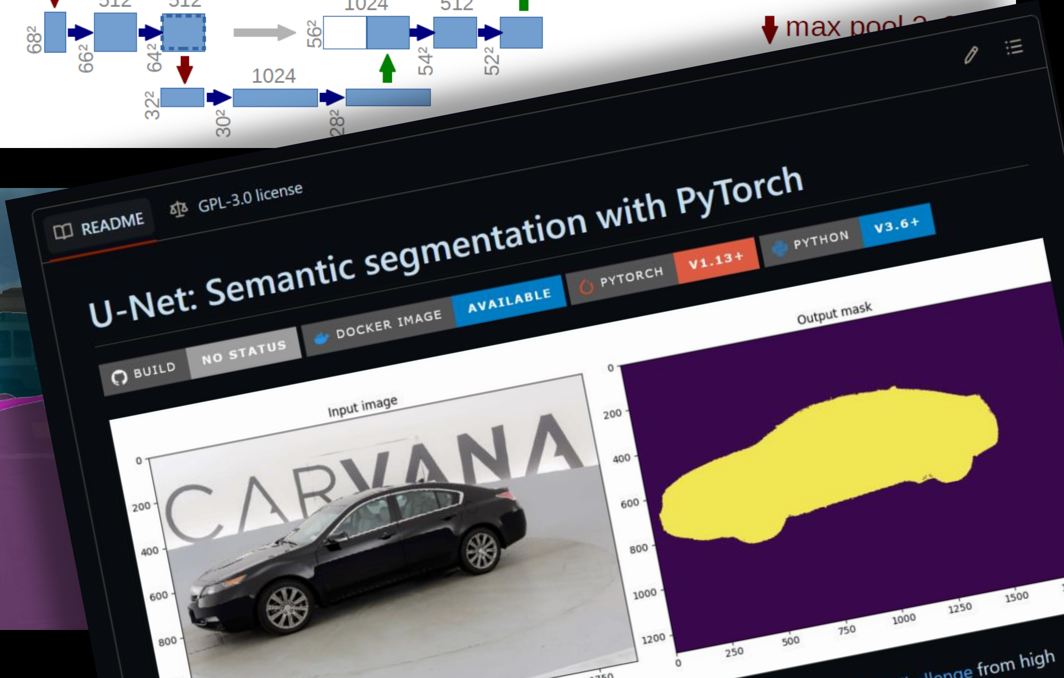


U-Net Architecture

- Encoder learns high-level scene features.
- Decoder upsamples features into a pixel-level mask.
- Skip connections preserve object boundaries.
- Output is a class label for every pixel.

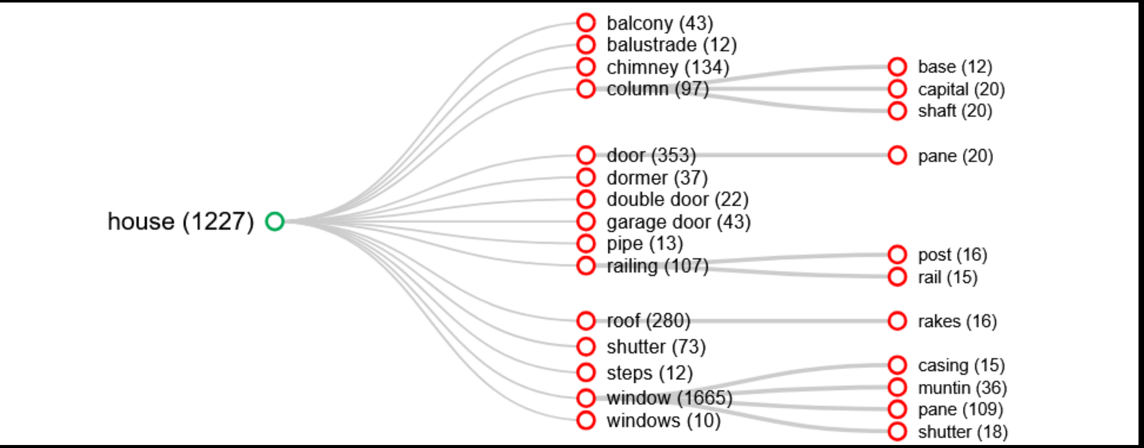




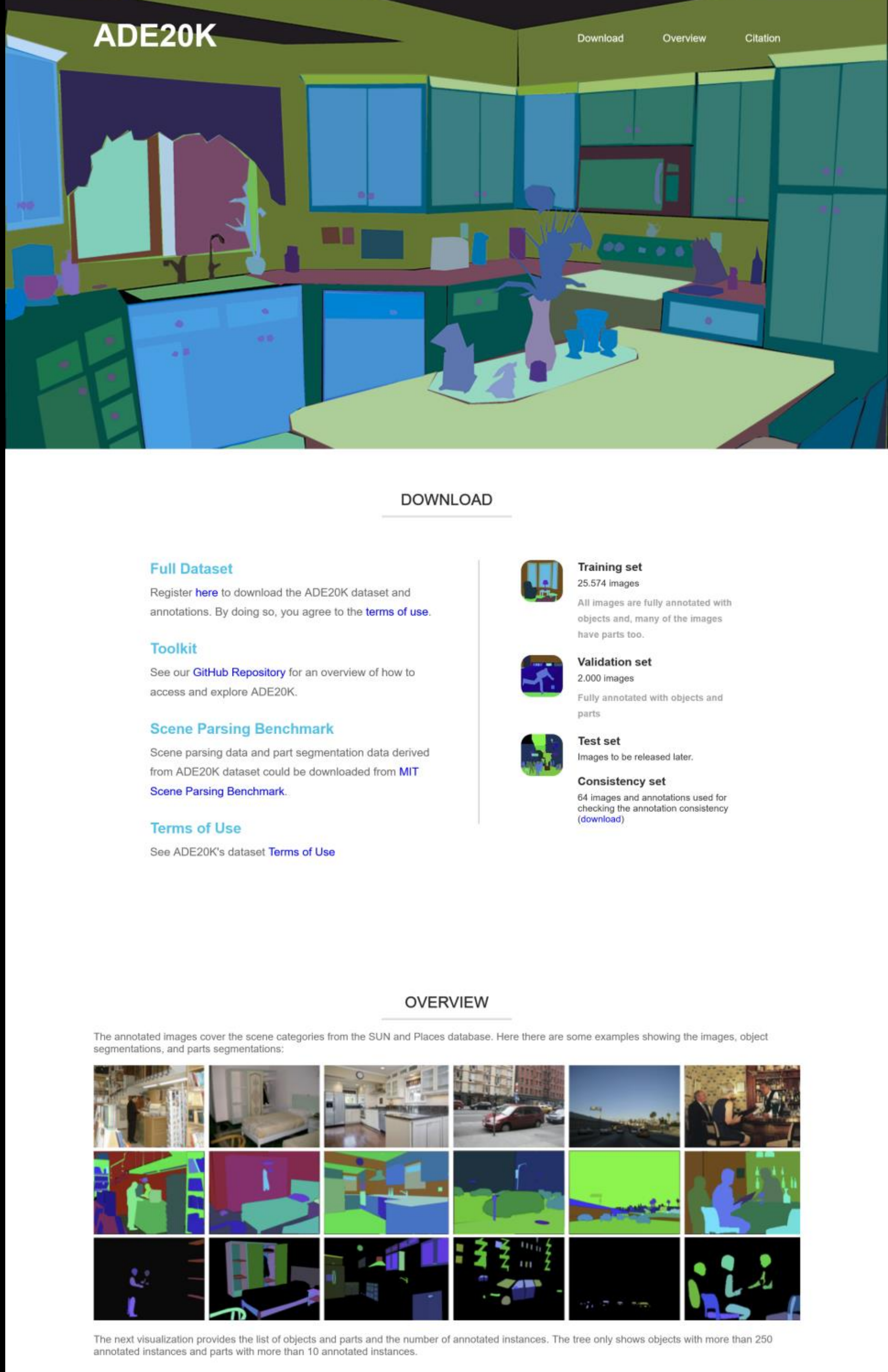


Data & Training

- Indoor images from classrooms and campus were taken.
- Public datasets such as ADE20K were used.
- These images were categorized into interior and outdoor scenes.
  - We used images that had the correct interior classes. (wall, floor, etc)



* this is an example of some of the classes in the ADE20K dataset.



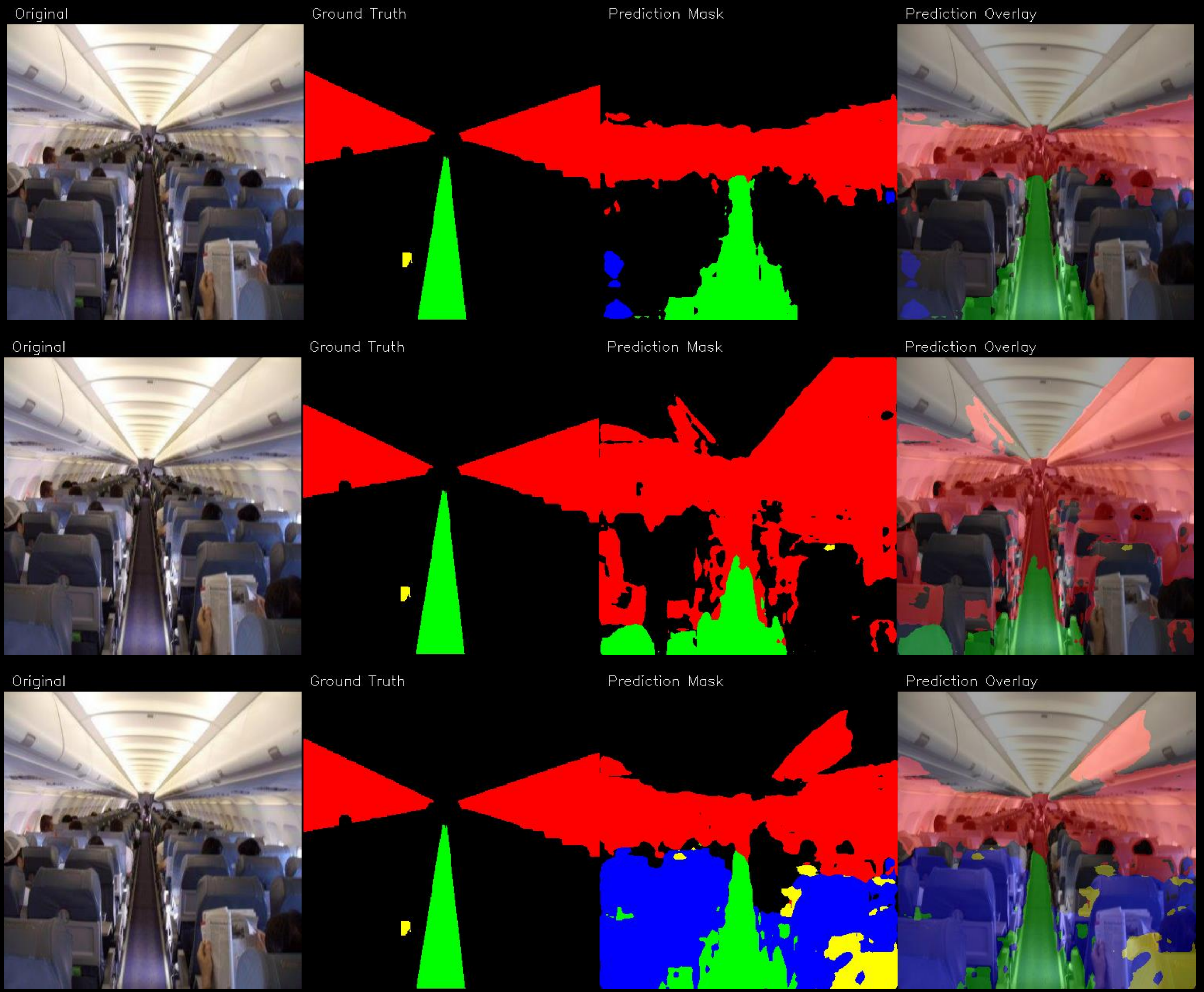
Training Overview

- Early Epochs introduce a lot of noise that is cleaned up over time.
- Predictions can be unstable.

Epoch 1

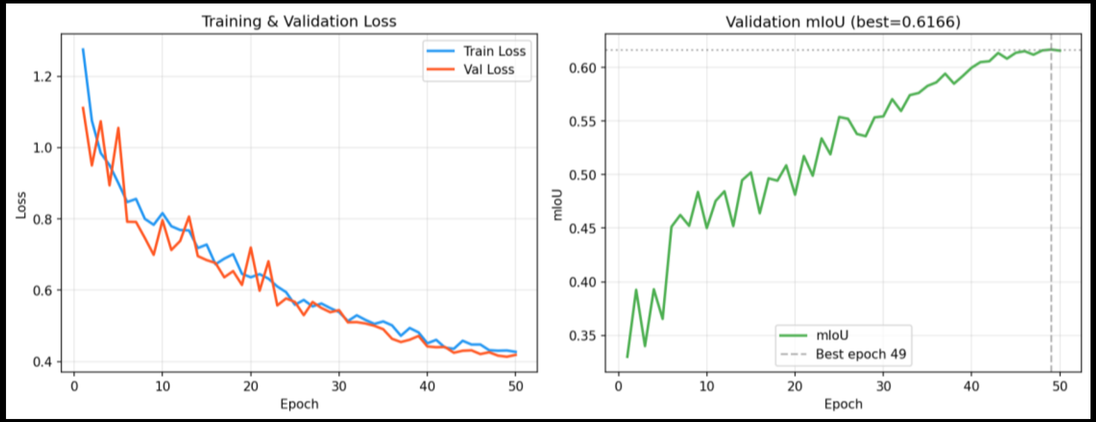
Epoch 2

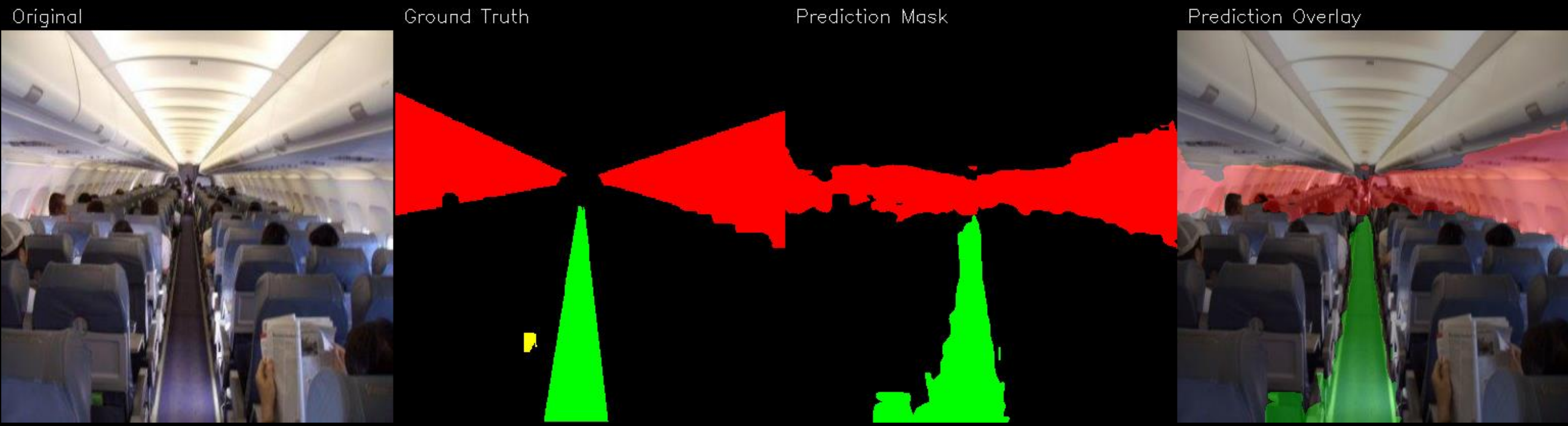
Epoch n...



5/15/2026

Training Overview

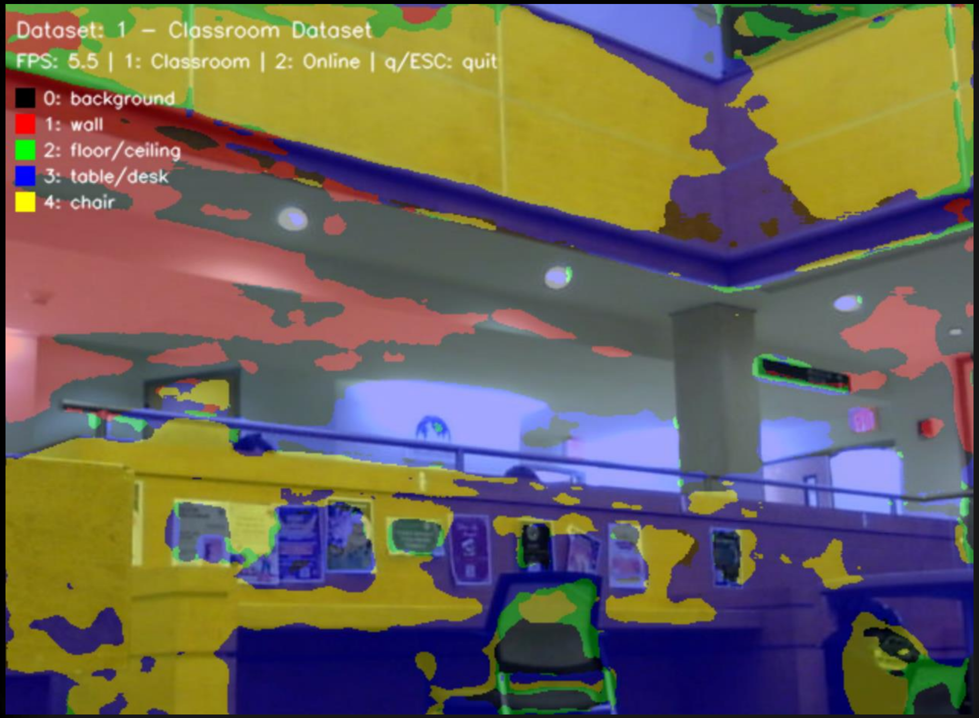




...Epoch ~50!

- By epoch 50, the model better captured large scene regions like walls and floors.

Results/Demo



* model using our (relatively small) dataset.

* model using the online ADE20k dataset, cleaned up to remove outdoor scenes.